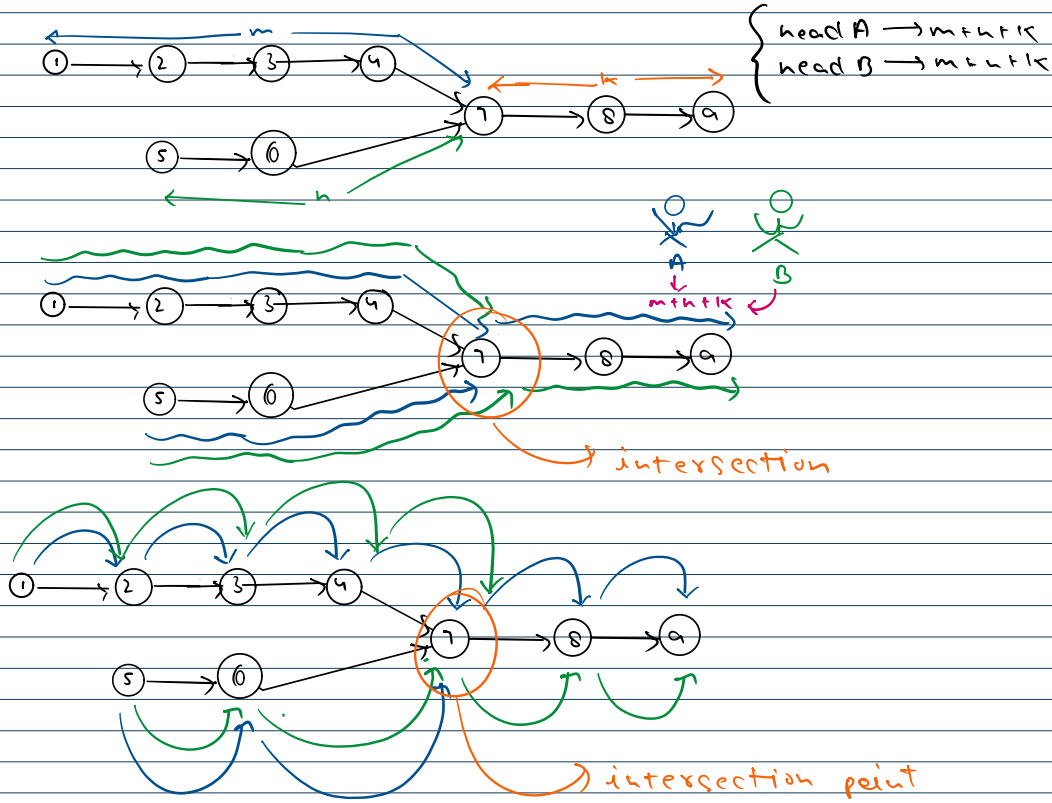
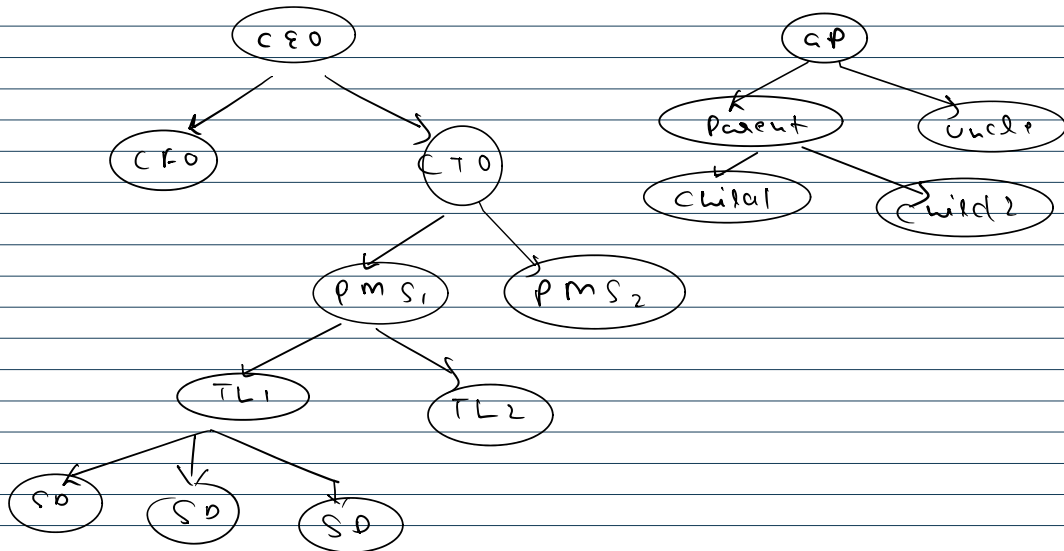


## Intersection of two Linked Lists

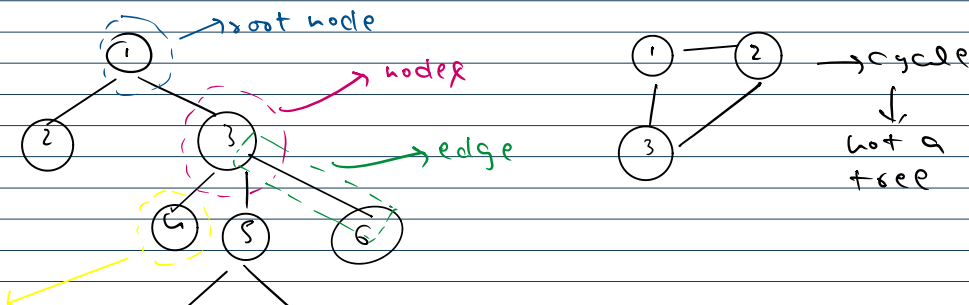


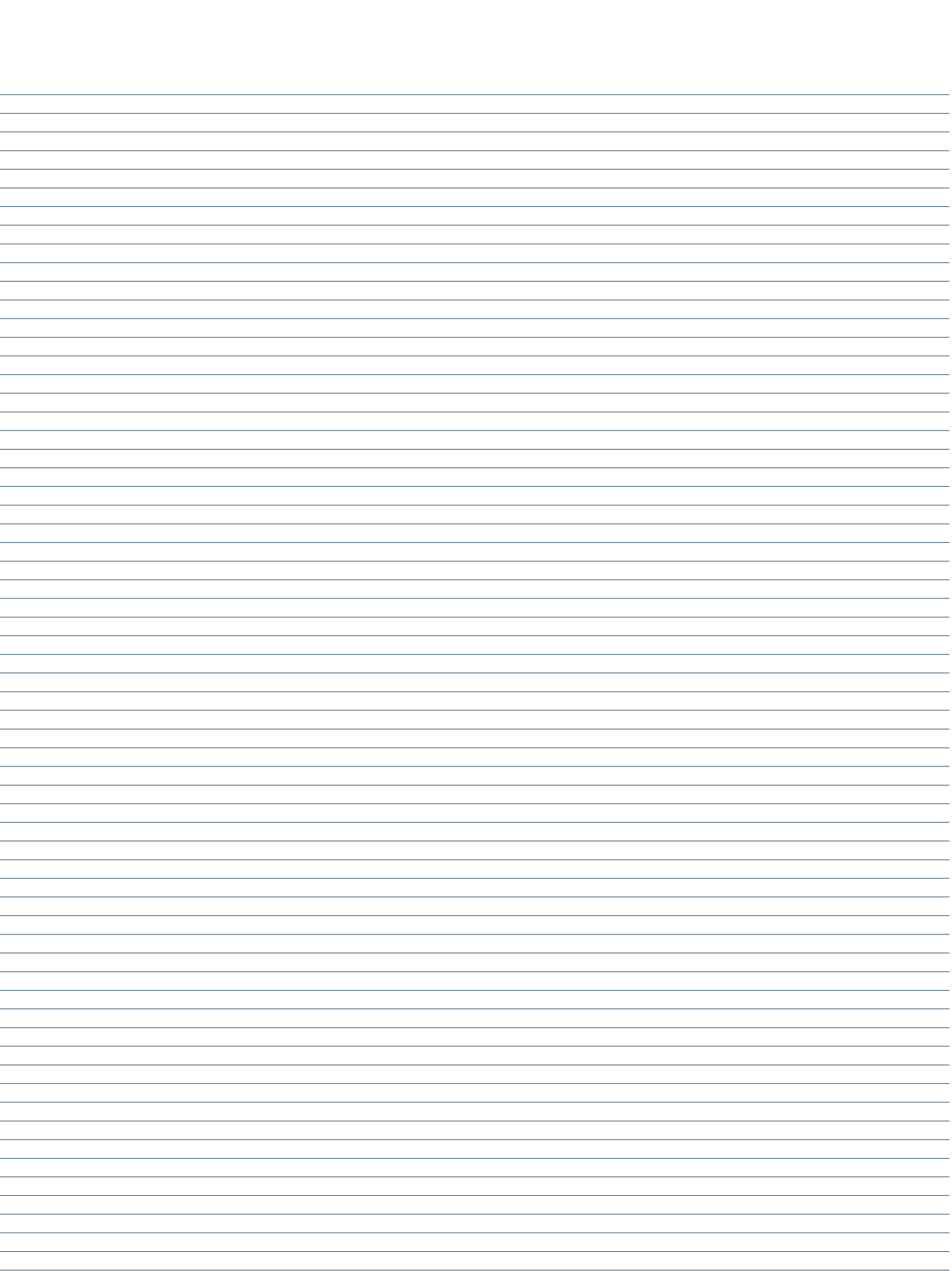
## Tree

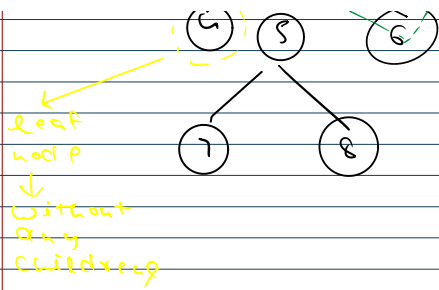
organisation  
chart  
(hierarchy)



It is a non-linear data structure that represents data in a hierarchical form.

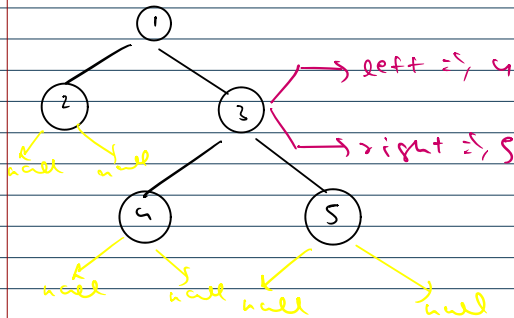




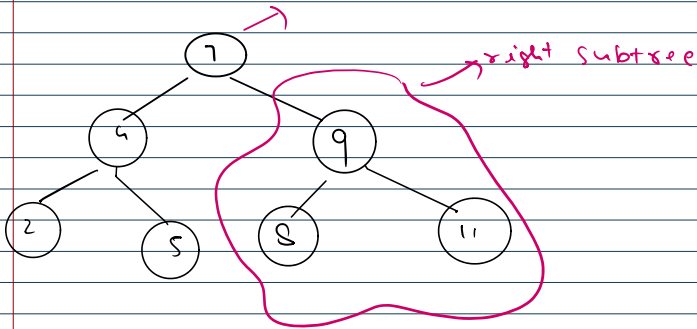


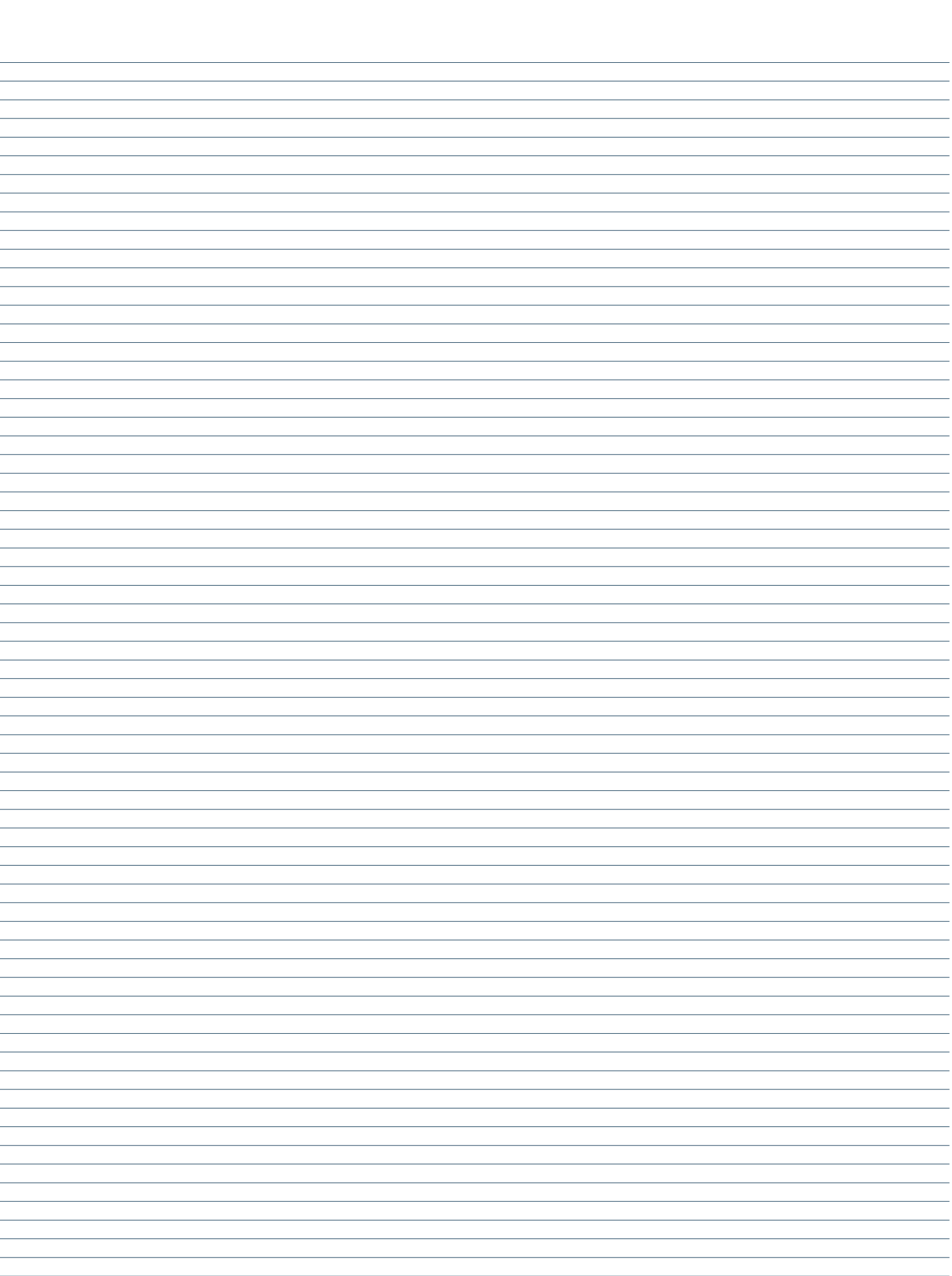
## Binary Tree

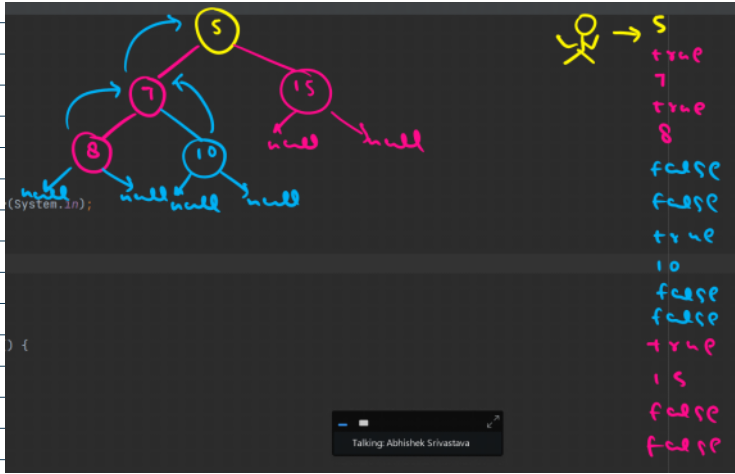
It is a special kind of tree where each node has at most  
two children



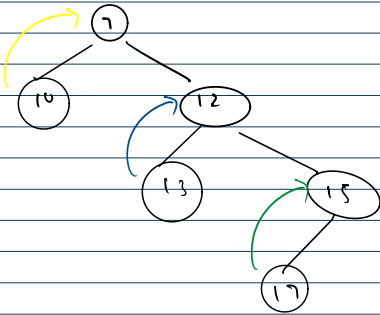
## Binary Search Tree (BST)





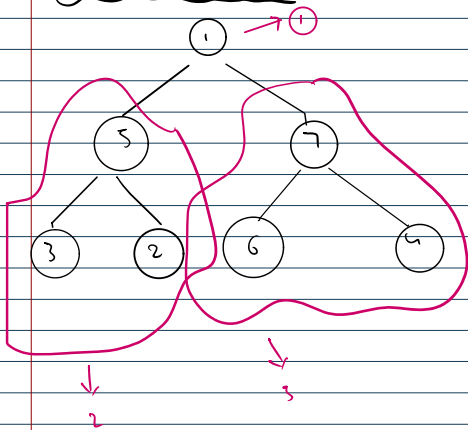


5  
 true  
 7  
 true  
 8  
 false  
 false  
 true  
 10  
 false  
 false  
 true  
 15  
 false  
 false



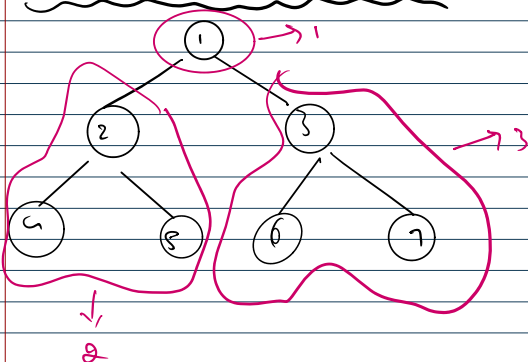
7  
 true  
 10  
 false  
 false  
 true  
 12  
 true  
 13  
 false  
 false  
 true  
 15  
 true  
 17  
 false  
 false  
 false

maximum value



$\text{Math.max}(\text{root.val},$   
 $\text{max}(\text{root.left}),$   
 $\text{max}(\text{root.right}))$

Search for an element

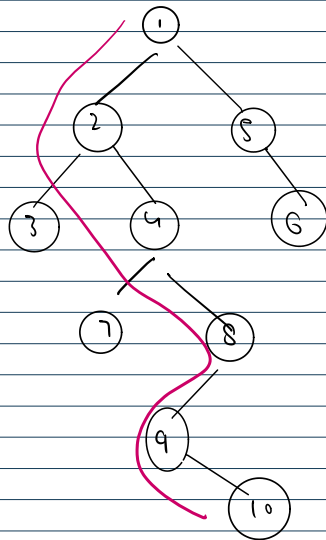


7 → search

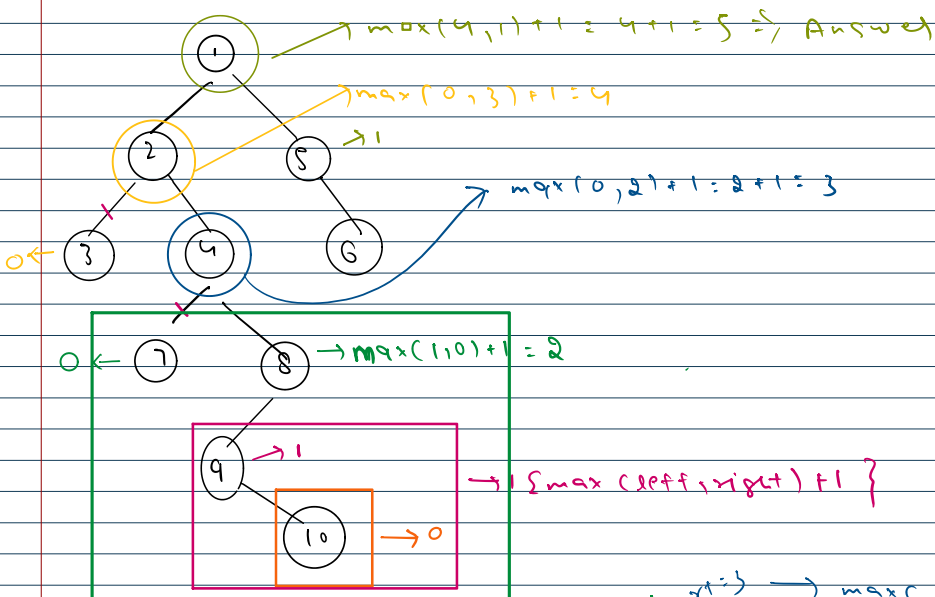
1, 2, 3 ⇒ if Found  
 ↓  
 true  
 ↓  
 false



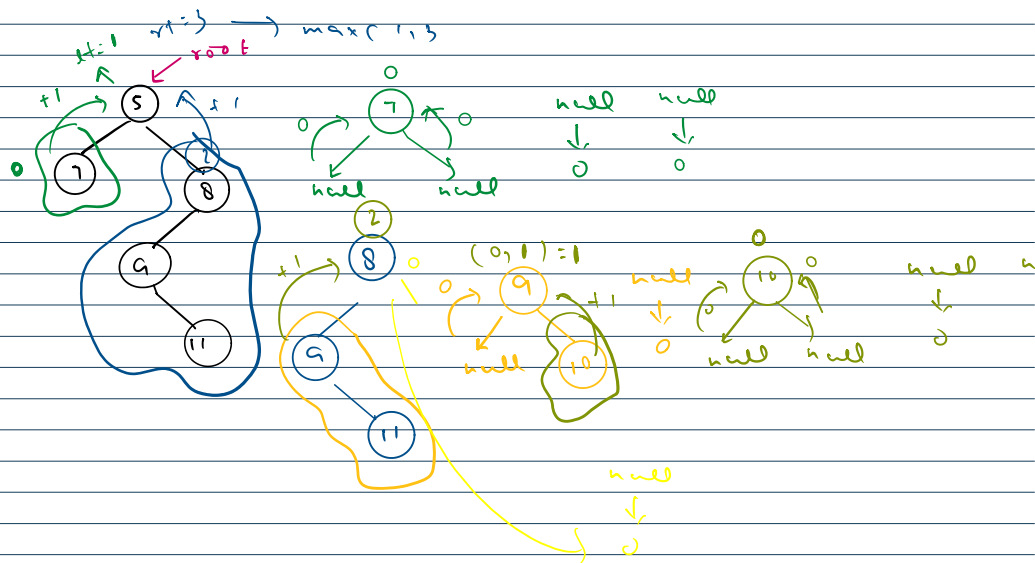
## Height of the tree



height = 5



```
private int height(Node root) {
    if (root == null) {
        return 0;
    }
    int lh = height(root.left);
    int rh = height(root.right);
    return Math.max(lh, rh) + 1;
}
```



## Traversal

Pre  
Post  
In  
Level

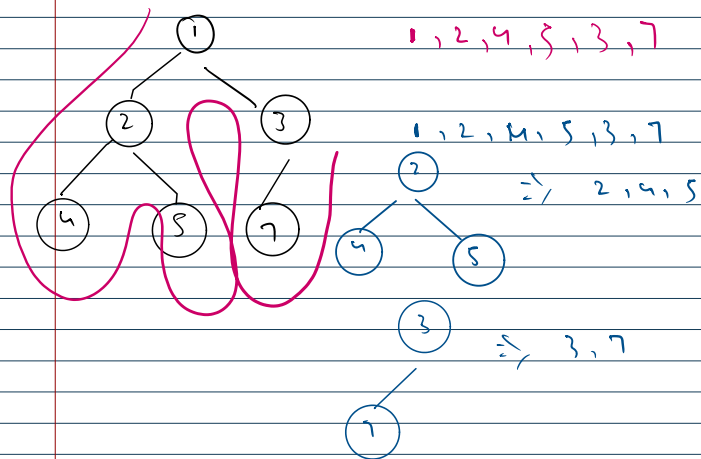
} order

20  
4  
5



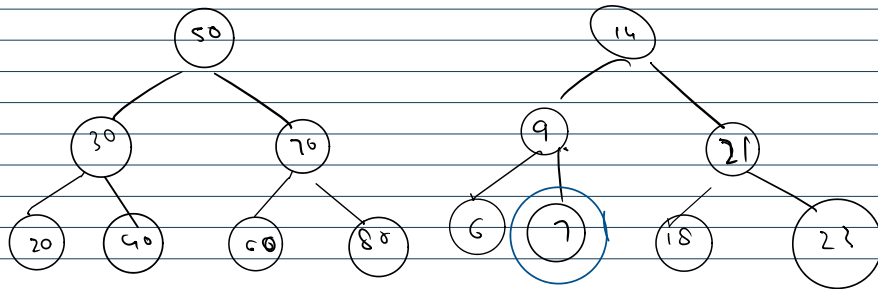
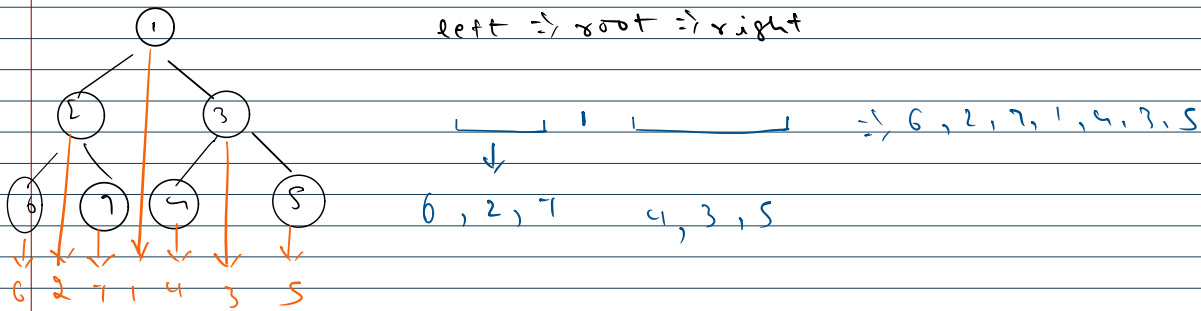
## Pre-order traversal

root  $\rightarrow$  left  $\rightarrow$  right

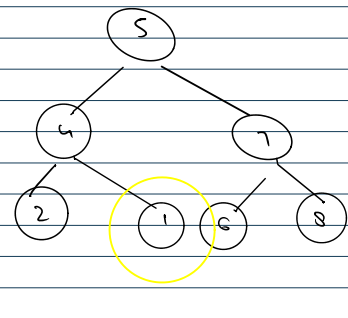


## Inorder traversal

left  $\Rightarrow$  root  $\Rightarrow$  right



6, 9, 7, 14, 18, 21, 23



2, 4, 1, 5, 6, 7, 8

## Post order Traversal

left  $\Rightarrow$  right  $\Rightarrow$  root

